Positive scalar curvature with point singularities

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Let $M^{n\geq 3}$ closed manifold, s.th. $\nexists$ PSC metric on M

↳ smooth Riemannian metric of positive scalar curvature

- M^n spin, $\hat{A}(M) \neq 0$ $\quad (n=4k)$

  K3 surface

- $M^n = T^n \# N$

- M^n spin, $\alpha(M) \neq 0 \in \mathbb{Z}/_2$ $\quad (n = 8k + \{1,2\})$

  (certain exotic spheres)

<u>Bourguignon:</u> If $\mathrm{scal}_g \geq 0$, then $\mathrm{Ric}_g \equiv 0$.

Theme: Riemannian manifolds with metric singularities

- L^∞-metric g on smooth mfd M ($\exists$ smooth metrics g_0, g_1 : $g_0 \le g \le g_1$)
  "$g \in L^\infty(M)$"
- g smooth outside a "sufficiently small" singular set $S \subseteq M$.
  $g \in C^\infty_{loc}(M \setminus S)$ ↳ $\operatorname{codim}(S \subseteq M) > 2$
- Study $\mathrm{scal}_g > c$ on $M \setminus S$.

Theorem (Li–Mantoulidis 2019): M^3... closed 3-manifold admitting no PSC metric
Let $S \subseteq M$ be finite, $g \in L^\infty(M) \cap C^\infty_{loc}(M \setminus S)$ and $\mathrm{scal}_g \ge 0$ on $M \setminus S$.
$\Rightarrow$ g is flat on $M \setminus S$ (and g extends smoothly to all of M.)

Conjecture (Schoen): M ... closed mfd, admits no (smooth) PSC metric.

Let $S \subseteq M$ submfd., $\operatorname{codim}(S \subset M) \geq 3$. Then:

- $\nexists\, g \in L^\infty(M) \cap C^\infty_{loc}(M \setminus S)$: $\operatorname{scal}_g > 0$ on $M \setminus S$. [Obstruction]
- $g \in L^\infty(M) \cap C^\infty_{loc}(M \setminus S)$, $\operatorname{scal}_g \geq 0$ on $M \setminus S$ $\Rightarrow$ $\operatorname{Ric}_g \equiv 0$ on $M \setminus S$. [Rigidity]

  Moreover, then g extends smoothly to M. [Extension]
  [after possibly changing the smooth atlas of M]

Some known cases:

- $\dim(M) = 3$ ✓ (Li-Mantoulidis 2019)
- $\dim(M) = 4$, M enlargeable, e.g. $M = T^4$ [obstruction] + [Rigidity] (Kazaras 2024)
- M enlargeable & spin (e.g. $M = T^4 \# N$, $S = \{p_1, \dots, p_N\}$) spin

  [obstruction] + [Rigidity] (Cecchini - Frenck - Z. 2024)

  [Extension] (Dai - Wang - Wang - Wei 2025)
- More "[obstruction] + [Rigidity]": Wang - Xie 2024, Shi - Wang - Wu - Zhu 2024

Alexander Trick:

$f: S^{n-1} \longrightarrow S^{n-1}$ diffeo.

$\rightsquigarrow \quad F: D^n \longrightarrow D^n$ homeomorphism

$x \longmapsto |x| f\left(\frac{x}{|x|}\right)$ smooth on $D^n \setminus \{0\}$

Note: g Riem. metric on $D^n \rightsquigarrow F^* g \in L^\infty(D^n) \cap C^\infty_{loc}(D^n \setminus \{0\})$

Example: $\Sigma \underset{\text{homeo}}{\cong} S^n \rightsquigarrow$

$$\begin{array}{ccl} \Sigma & = & D^n \cup_f D^n \qquad f: S^{n-1} \circlearrowleft \text{ Diffeo} \\ \downarrow & & \qquad \downarrow \mathrm{id} \cup F \\ S^n & = & D^n \cup_{\mathrm{id}} D^n \end{array}$$

Thm (Cecchini - Frenck - Z. 2024) Any exotic sphere $\Sigma^{n \geq 7}$ admits:

- an L^∞-metric g smooth away from $p \in \Sigma$
- a homeo $F: \Sigma^n \to S^n$, diffeo on $\Sigma^n \setminus \{p\}$,

  $F^* g_0 = g$ on $\Sigma^n \setminus \{p\}$

Remarks: • $\exists$ exotic spheres which do not admit PSC

$\rightsquigarrow$ counterexamples to [obstruction] !

[Extension] ?

Thm (CFZ 2024): Let $\Sigma^{n\geq 7}$ be an exotic sphere, $T_\Sigma := T^n \# \Sigma$.

$\Longrightarrow$ T_Σ admits an L^∞-metric smooth & flat away from a point $p \in T_\Sigma$.

But: T_Σ does not admit any smooth metric of non-negative scalar curvature

Theorem A [Cecchini - French - Z. 2024]

[not homeomorphic to PSC manifolds]

$\forall\ n \geq 8 \quad \exists$ closed smooth M^n without smooth PSC

$\exists g \in L^\infty(M) \cap C^\infty_{loc}(M \setminus S)$, $\operatorname{codim}(S \subseteq M) \geq 8$,

$\operatorname{scal}_g > 0$ on $M \setminus S$.

$\rightsquigarrow$ More counterexamples to [Obstruction] ?!

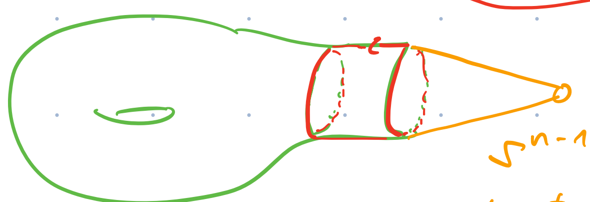
Ingredients:

① PSC π - π - principle [Gromov - Lawson, Gajer] $\rightsquigarrow$ $M^{n \geq 6}$ 1-connected spin $\Rightarrow$ $\exists$ PSC metric $\tilde{g}$ on $M \setminus D^n$ with a collar.

$S^{n-1} \times [-\varepsilon, 0]$

$\tilde{g} = h + dt^2$ $\in \mathcal{R}^+(S^{n-1})$

② Glue cone to $\tilde{g}$ as in ①



$S^{n-1} \times [\varepsilon, r) \cong D^n \setminus \{0\}$

$\left(\frac{r-t}{\lambda}\right)^2 h + dt^2 \in L^\infty(D^n)$

$\rightsquigarrow \exists g \in L^\infty(M) \cap C^\infty_{loc}(M \setminus \{0\})$

$\operatorname{scal}_g > 0$ on $M \setminus \{0\}$

③ Spin obstructions: $\exists$ 1-connected manifolds without PSC if $4 \leq n \equiv 0,1,2,4 \mod 8$

e.g. • $M = K3 \times K3 \times \dots \times K3$ for $n \equiv 0 \mod 4$ $[\hat{A}(M) \neq 0]$

• Exotic spheres Σ^n, $9 \leq n \equiv 1,2 \mod 8$, $\alpha(\Sigma) \neq 0 \in \mathbb{Z}/2\mathbb{Z}$.

By ① & ②, any such M with $n \geq 8$ is a counterexample to [obstruction] with $S = \{0\}$.

Examples for other $n \geq 8$ are obtained by $\cdot \times T^k$

• Remark: Many of the examples from Theorem A are not homeomorphic to manifolds admitting PSC. (e.g. $K3 \times K3 \times \dots \times K3 \times T^n$.)

$g \in L^\infty(M) \cap C^\infty_{loc}(M \setminus S)$ + conditions PSC on $M \setminus S$ $\overset{\text{blow up}}{\rightsquigarrow}$ procedure Complete PSC on $M \setminus S$

Sometimes there are top. obstructions to this.
e.g. $T^4 \setminus \{p_1, \dots, p_N\}$.

Local approximation of L^∞-metrics?

Theorem (Burkhard-Guim 2024): $\forall \alpha > 2$, $n \geq 3$ $\exists \bar{\varepsilon} = \bar{\varepsilon}(\alpha, n)$:

Let g L^∞-metric on $\mathbb{R}^n$, $\|g - \delta\|_{L^\infty} < \bar{\varepsilon}$ s.th.

- $g \in C^\infty_{loc}(\mathbb{R}^n \setminus S)$, where S has finite $(n-\alpha)$-dim. Minkowski content
- $\mathrm{scal}_g \geq 0$ on $\mathbb{R}^n \setminus S$

$\Longrightarrow$ $\exists (g_t)_{t>0}^{\in C^\infty(\mathbb{R}^n)}$: $\mathrm{scal}_{g_t} \geq 0$, $g_t \xrightarrow[t \searrow 0]{C^\infty_{loc}(\mathbb{R}^n \setminus S)} g$.

Q.: Do we need $\|g - \delta\|_{L^\infty} \ll 1$?

Recall: $\exists M$ closed, no PSC : $\exists \tilde{g}$ collared PSC metric on $M^n \setminus D^n$

i.e. $\tilde{g} = h + dt^2$ near $\partial(M^n \setminus D^n)$

$\rightsquigarrow$ h PSC metric on S^{n-1}.

$\rightsquigarrow$ h does not extend to a collared PSC metric on D^n [otherwise $M = M \setminus D^n \cup_{S^{n-1}} D^n$ would admit PSC]

Theorem B (CFZ 2024) For $n \equiv 0,1,2,4 \mod 8$, $\exists\, g \in L^\infty(\mathbb{R}^n)$

s.th.:

- $\Lambda^{-1}\delta \le g \le \Lambda\delta$ for some $\Lambda>0$
- $g \in C^\infty_{loc}(\mathbb{R}^n \setminus \{0\})$
- $scal_g \ge c\,|x|^{-2}$ for some $c>0$.

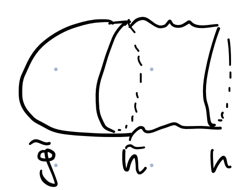
But: $\nexists\, (g_\epsilon)_{\epsilon>0} \subset C^\infty$: $scal_{g_\epsilon} \ge 0$, $g_\epsilon \xrightarrow[\epsilon \searrow 0]{C^\infty_{loc}(\mathbb{R}\setminus\{0\})} g$.

Proof idea: Take $g = \left(\frac{t}{\lambda}\right)^2 h + dt^2$ for suitable $\lambda > 0$.

If $\exists$ approximation $g_\epsilon \xrightarrow{C^\infty_{loc}(\mathbb{R}^n\setminus\{0\})} g$, $scal_{g_\epsilon} \ge 0$, then h is C^∞-close to a metric

$\tilde h$ on S^{n-1} which extends to a collared PSC metric $\tilde g$ on D^n. But then

h itself extends to a collared PSC metric, ↯.



Remarks

- $g \in W^{2,\frac{p}{2}}_{loc} \subset W^{1,p}_{loc}$ for all $p < n$ [in all our examples]

- $Scal_g$ is defined "globally" in the weak sense as an $L^{p/2}$-function, $p<n$

- In contrast, there are Ricci-deTurck flow approximation results for $g \in L^\infty \cap W^{1,n}$ (Lamm-Simon, Chu-Lee 2022)